Operating Systems Model Practice Paper

Operating Systems - Demo paper - 2025 edition - 50 marks

Self-created practice paper for NoteNest. It does not reproduce any official university question paper.

Section A - Short answers (5 x 4 marks)

Q1. Distinguish between a process and a thread. **[4 marks]**

Q2. What information is stored in a Process Control Block? **[4 marks]**

Q3. Define turnaround time, waiting time and response time. **[4 marks]**

Q4. What is a page fault? Describe the steps the OS takes to handle one. **[4 marks]**

Q5. State the four necessary conditions for deadlock. **[4 marks]**

Section B - Long answers (2 x 15 marks)

Q6. Processes P1(arrival 0, burst 6), P2(1, 4), P3(2, 2) are scheduled with Round Robin, quantum 2. Draw the Gantt chart and compute the average waiting and turnaround times. **[15 marks]**

Q7. For the reference string 7 0 1 2 0 3 0 4 2 3 with 3 frames, count the page faults under FIFO and LRU. Which performs better here and why? **[15 marks]**

Answer hints

- Q3: turnaround = completion - arrival; waiting = turnaround - burst; response = first CPU time - arrival.
- Q7: FIFO gives 9 faults on this string; LRU gives 8 - verify by drawing the frame table.